

✓ Telco Customer Churn Analysis and Prediction

By Atharva Salitri

a. Project Overview

This project focuses on analyzing customer churn in a telecommunications company. Customer churn, the phenomenon of customers discontinuing their service, is a critical business problem for Telco companies as it directly impacts revenue and profitability. By understanding the factors contributing to churn, the company can develop targeted strategies to retain valuable customers.

b. Dataset

The dataset used for this analysis is the `Telco Customer Churn` dataset, which contains information about 7,032 customers. It includes 20 features covering demographic information, services subscribed to, contract details, and payment information, along with a `Churn` target variable indicating whether a customer left the company.

c. Problem Statement

The primary goal is to identify factors influencing customer churn and build predictive models to accurately forecast which customers are likely to churn. This will enable the Telco company to proactively intervene with retention strategies.

d. Objectives

- **Data Understanding and Cleaning:** Explore the dataset, identify data quality issues, and preprocess the data for analysis.
- **Exploratory Data Analysis (EDA) & Visualization:** Uncover patterns, relationships, and key drivers of churn through comprehensive data visualization.
- **Machine Learning Modeling:** Train and evaluate several classification models to predict customer churn.
- **Business Insights & Recommendations:** Translate analytical findings into actionable business strategies to reduce churn.

✓ Code and Analysis:

✓ 1. Importing Libraries and Loading Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('/content/Telco-Customer-Churn.csv')
```

✓ 2. Data Understanding

Let's get a basic understanding of the dataset by looking at the first few rows, checking data types, and summary statistics.

```
print("First 5 rows of the dataset:")
display(df.head())

print("\nDataset Info:")
df.info()

print("\nDescriptive Statistics:")
```

```
display(df.describe(include='all'))  
print("\nDataset Shape:", df.shape)
```

First 5 rows of the dataset:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DS
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DS
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DS
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DS
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber opt

5 rows × 21 columns

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null	Count	Dtype
0	customerID	7043	non-null	object
1	gender	7043	non-null	object
2	SeniorCitizen	7043	non-null	int64
3	Partner	7043	non-null	object
4	Dependents	7043	non-null	object
5	tenure	7043	non-null	int64
6	PhoneService	7043	non-null	object
7	MultipleLines	7043	non-null	object
8	InternetService	7043	non-null	object
9	OnlineSecurity	7043	non-null	object
10	OnlineBackup	7043	non-null	object
11	DeviceProtection	7043	non-null	object
12	TechSupport	7043	non-null	object
13	StreamingTV	7043	non-null	object
14	StreamingMovies	7043	non-null	object
15	Contract	7043	non-null	object
16	PaperlessBilling	7043	non-null	object
17	PaymentMethod	7043	non-null	object
18	MonthlyCharges	7043	non-null	float64
19	TotalCharges	7043	non-null	object
20	Churn	7043	non-null	object

dtypes: float64(1), int64(2), object(18)

memory usage: 1.1+ MB

Descriptive Statistics:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService
count	7043	7043	7043.000000	7043	7043	7043.000000	7043	7043	
unique	7043	2	NaN	2	2	NaN	2	3	
top	3186-AJIEK	Male	NaN	No	No	NaN	Yes	No	
freq	1	3555	NaN	3641	4933	NaN	6361	3390	
mean	NaN	NaN	0.162147	NaN	NaN	32.371149	NaN	NaN	
std	NaN	NaN	0.368612	NaN	NaN	24.559481	NaN	NaN	
min	NaN	NaN	0.000000	NaN	NaN	0.000000	NaN	NaN	
25%	NaN	NaN	0.000000	NaN	NaN	9.000000	NaN	NaN	
50%	NaN	NaN	0.000000	NaN	NaN	29.000000	NaN	NaN	
75%	NaN	NaN	0.000000	NaN	NaN	55.000000	NaN	NaN	
max	NaN	NaN	1.000000	NaN	NaN	72.000000	NaN	NaN	

11 rows × 21 columns

Dataset Shape: (7043, 21)

3. Data Cleaning and Preprocessing

Based on the initial inspection, we need to address a few things:

- The `TotalCharges` column is of object type, but should be numeric. We'll convert it and handle any errors.
- Check for and handle missing values.
- Drop the `customerID` column as it's not relevant for prediction.

```
# Convert 'TotalCharges' to numeric. Coerce errors will turn non-numeric values into NaN
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Check for missing values after conversion
print("Missing values after converting TotalCharges:")
print(df.isnull().sum())

# Drop rows with missing TotalCharges (which were ' ' in the original dataset)
df.dropna(inplace=True)
print("\nDataset Shape after dropping nulls:", df.shape)

# Drop customerID column
df.drop('customerID', axis=1, inplace=True)
print("\nDropped 'customerID' column.")

# Display data types after cleaning
print("\nData types after cleaning:")
df.info()
```

Missing values after converting TotalCharges:

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	11
Churn	0

dtype: int64

Dataset Shape after dropping nulls: (7032, 21)

Dropped 'customerID' column.

Data types after cleaning:

```
<class 'pandas.core.frame.DataFrame'>
```

Index: 7032 entries, 0 to 7042

Data columns (total 20 columns):

#	Column	Non-Null Count	Dtype
0	gender	7032 non-null	object
1	SeniorCitizen	7032 non-null	int64
2	Partner	7032 non-null	object
3	Dependents	7032 non-null	object
4	tenure	7032 non-null	int64
5	PhoneService	7032 non-null	object
6	MultipleLines	7032 non-null	object
7	InternetService	7032 non-null	object
8	OnlineSecurity	7032 non-null	object
9	OnlineBackup	7032 non-null	object
10	DeviceProtection	7032 non-null	object
11	TechSupport	7032 non-null	object
12	StreamingTV	7032 non-null	object
13	StreamingMovies	7032 non-null	object
14	Contract	7032 non-null	object
15	PaperlessBilling	7032 non-null	object

```

16 PaymentMethod      7032 non-null    object
17 MonthlyCharges      7032 non-null    float64
18 TotalCharges        7032 non-null    float64
19 Churn               7032 non-null    object
dtypes: float64(2), int64(2), object(16)
memory usage: 1.1+ MB

```

4. Target Variable Analysis (Churn)

Let's analyze the distribution of our target variable, `Churn`, to understand the class imbalance.

```

print("Distribution of Churn:")
print(df['Churn'].value_counts())
print("\nPercentage of Churn:")
print(df['Churn'].value_counts(normalize=True) * 100)

plt.figure(figsize=(6, 4))
sns.countplot(x='Churn', data=df, palette='viridis')
plt.title('Distribution of Churn (Yes/No)')
plt.xlabel('Churn')
plt.ylabel('Number of Customers')
plt.show()

```

```

Distribution of Churn:
Churn
No      5163
Yes     1869
Name: count, dtype: int64

```

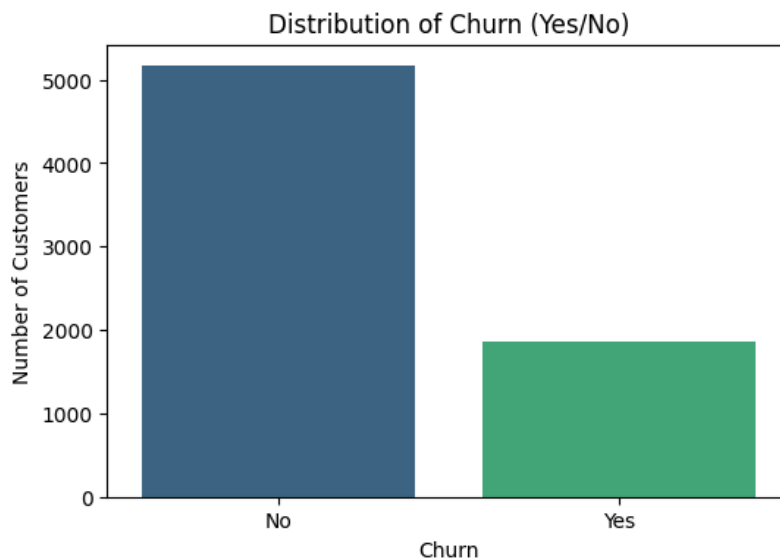
```

Percentage of Churn:
Churn
No      73.421502
Yes     26.578498
Name: proportion, dtype: float64
/tmp/ipykernel_530/1267961077.py:7: FutureWarning:

```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable

```
sns.countplot(x='Churn', data=df, palette='viridis')
```



5. Data Visualization - Exploring Categorical Features vs. Churn

Let's visualize the relationship between categorical features and the 'Churn' variable using count plots.

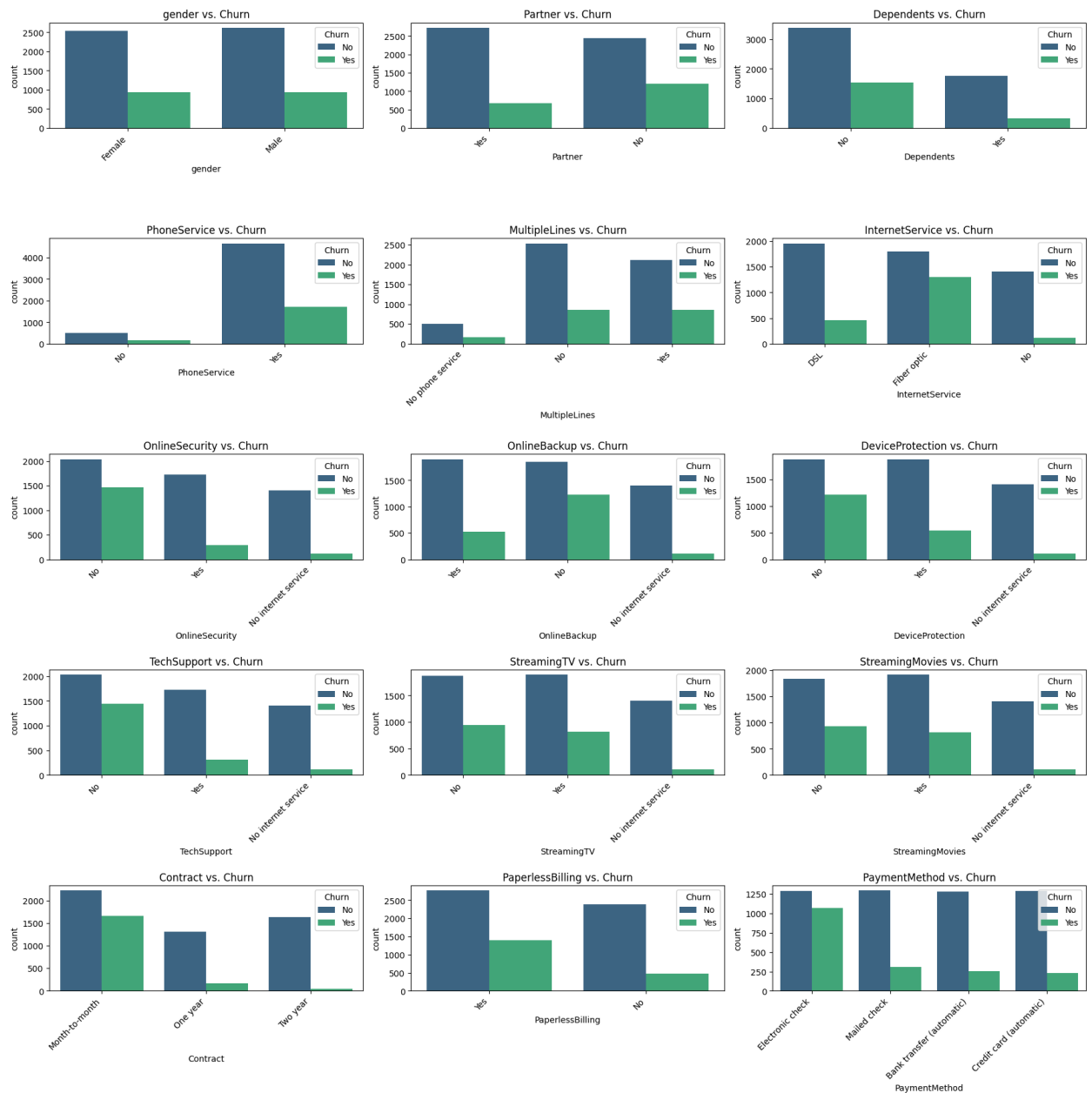
```

categorical_cols = df.select_dtypes(include='object').columns.tolist()
categorical_cols.remove('Churn') # Exclude the target variable

plt.figure(figsize=(18, 20))
for i, col in enumerate(categorical_cols):
    plt.subplot(6, 3, i + 1) # Adjust subplot grid as needed

```

```
sns.countplot(x=col, hue='Churn', data=df, palette='viridis')
plt.title(f'{col} vs. Churn')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



```
print('Churn percentages by SeniorCitizen:')
display(df.groupby('SeniorCitizen')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by Partner:')
display(df.groupby('Partner')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by Dependents:')
display(df.groupby('Dependents')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by InternetService:')
display(df.groupby('InternetService')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by OnlineSecurity:')
display(df.groupby('OnlineSecurity')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by TechSupport:')
display(df.groupby('TechSupport')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by Contract:')
display(df.groupby('Contract')['Churn'].value_counts(normalize=True).unstack() * 100)
```

```
print('\nChurn percentages by PaperlessBilling:')
display(df.groupby('PaperlessBilling')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages by PaymentMethod:')
display(df.groupby('PaymentMethod')['Churn'].value_counts(normalize=True).unstack() * 100)

print('\nChurn percentages for Gender:')
display(df.groupby('gender')['Churn'].value_counts(normalize=True).unstack() * 100)
```


Churn percentages by SeniorCitizen:

	Churn	No	Yes
SeniorCitizen			
0	76.349745	23.650255	
1	58.318739	41.681261	

Churn percentages by Partner:

	Churn	No	Yes
Partner			
No	67.023908	32.976092	
Yes	80.282935	19.717065	

Churn percentages by Dependents:

	Churn	No	Yes
Dependents			
No	68.720860	31.279140	
Yes	81.466795	18.531205	

Insights from Categorical Feature Visualization

- Gender:** There is a significant difference in churn rates between male (**26.63%** churn) and female (**26.53%** churn) customers.
- SeniorCitizen:** Senior citizens (**41.68%** churn) show a significantly higher churn rate compared to non-senior citizens (**23.65%** churn).
- Partner, Dependents:** Customers without partners (**32.98%** churn) or dependents (**31.39%** churn) tend to churn more, suggesting that single customers or those without family responsibilities might be less tied to the service.
- PhoneService:** While the vast majority of customers have phone service, customers *without* phone service have a **23.09%** churn rate, slightly lower than those with phone service (**26.92%** churn). This difference is not as stark as other features.
- MultipleLines:** Customers with multiple lines (**28.69%** churn) churn slightly more than those with a single line (**24.99%** churn), but the difference is not extremely pronounced.
- InternetService:** Customers with Fiber optic internet service have a significantly higher churn rate (**41.88%** churn) compared to DSL (**18.96%** churn) or no internet service (**7.49%** churn). This could indicate dissatisfaction with Fiber optic service quality or higher pricing.
- OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies:** Customers who do *not* have these services (e.g., Online Security: **42.14%** churn; Tech Support: **42.06%** churn) show considerably higher churn rates. This suggests that these supplementary services act as retention factors. Conversely, streaming services (TV and Movies) don't show as strong a relationship, with churn being somewhat balanced.
- Contract:** Month-to-month contract customers have a much higher churn rate (**42.71%** churn) than those with one-year (**11.25%** churn) or two-year contracts (**3.66%** churn), indicating that longer-term contracts are effective for retention.
- PaperlessBilling:** Customers with paperless billing (**33.73%** churn) have a higher churn rate compared to those without (**16.33%** churn), which might suggest they are more tech-savvy and willing to switch providers.
- PaymentMethod:** Electronic check users have the highest churn rate (**45.33%** churn), which could be due to ease of switching or specific demographics using this payment method, compared to other methods like Bank transfer (automatic) at **16.71%** churn, Credit card (automatic) at **16.01%** churn, and Mailed check at **20.97%** churn.

6. Data Visualization - Exploring Numerical Features vs. Churn

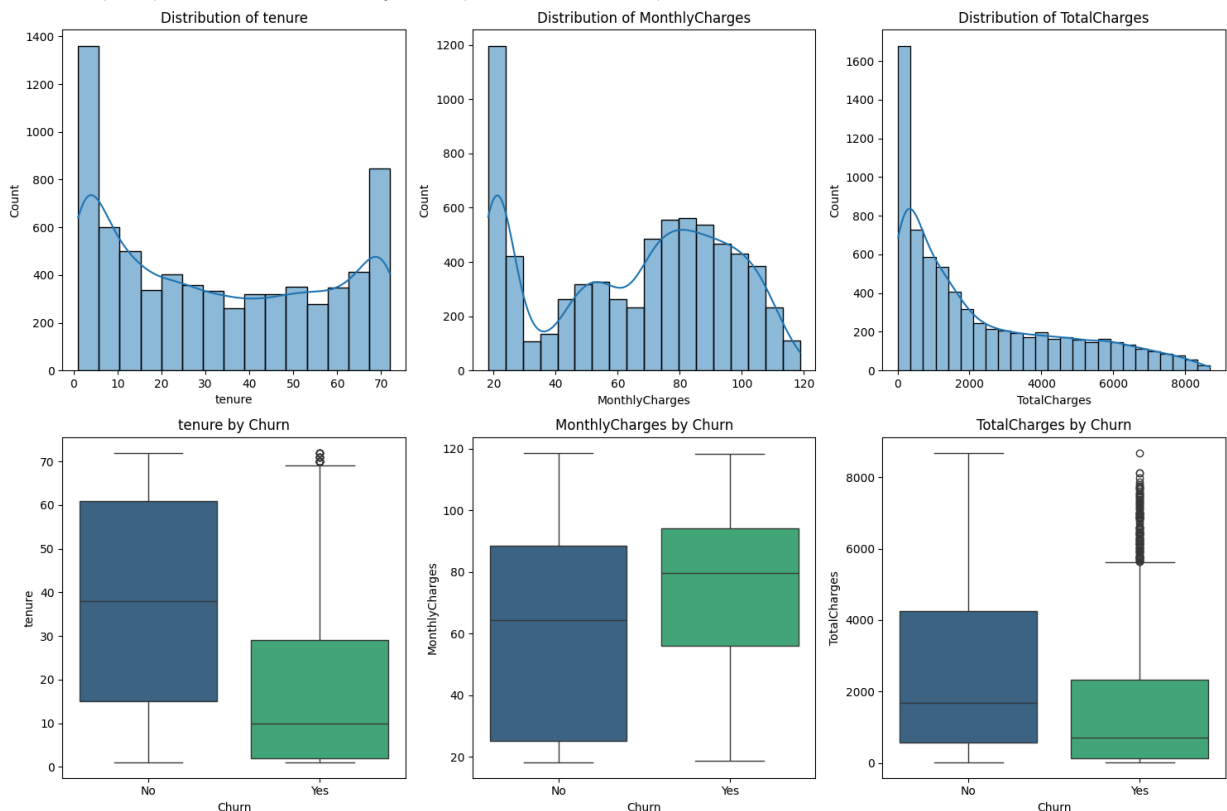
Let's visualize the distribution of numerical features and their relationship with 'Churn'.

```
numerical_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']

plt.figure(figsize=(15, 10))
for i, col in enumerate(numerical_cols):
    plt.subplot(2, 3, i + 1) # Histogram for overall distribution
    sns.histplot(data=df, x=col, kde=True, palette='viridis')
    plt.title(f'Distribution of {col}')
```

```
plt.subplot(2, 3, i + 4) # Box plot for churn comparison
sns.boxplot(data=df, x='Churn', y=col, palette='viridis')
plt.title(f'{col} by Churn')
plt.tight_layout()
plt.show()

/tmp/ipykernel_530/1562429310.py:6: UserWarning: Ignoring `palette` because no `hue` variable has been assigned
sns.histplot(data=df, x=col, kde=True, palette='viridis')
/tmp/ipykernel_530/1562429310.py:10: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable manually to `hue` instead.
sns.boxplot(data=df, x='Churn', y=col, palette='viridis')
/tmp/ipykernel_530/1562429310.py:6: UserWarning: Ignoring `palette` because no `hue` variable has been assigned
sns.histplot(data=df, x=col, kde=True, palette='viridis')
/tmp/ipykernel_530/1562429310.py:10: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable manually to `hue` instead.
sns.boxplot(data=df, x='Churn', y=col, palette='viridis')
/tmp/ipykernel_530/1562429310.py:6: UserWarning: Ignoring `palette` because no `hue` variable has been assigned
sns.histplot(data=df, x=col, kde=True, palette='viridis')
/tmp/ipykernel_530/1562429310.py:10: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable manually to `hue` instead.
```



Insights from Numerical Feature Visualization

- **Tenure:** Customers with lower `tenure` (newer customers) have a much higher churn rate. As `tenure` increases, the likelihood of churn decreases significantly. This indicates that long-term customers are more loyal.
- **MonthlyCharges:** Customers with higher `MonthlyCharges` tend to churn more. This suggests that expensive plans might be a contributing factor to churn.
- **TotalCharges:** The distribution of `TotalCharges` shows that churning customers often have lower `TotalCharges` compared to non-churning customers, which aligns with the observation that new customers churn more. Non-churning customers show a wider range and higher median `TotalCharges`.

Advanced Data Visualisations:

6.1 Overall Churn Distribution

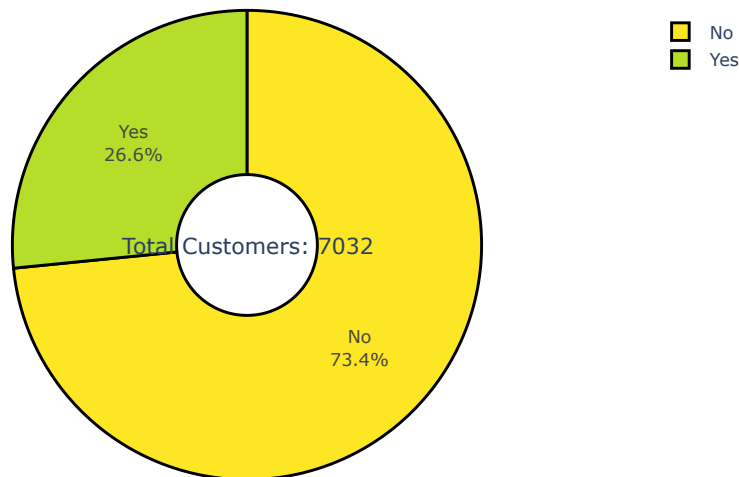
```
churn_counts = df['Churn'].value_counts()
churn_percentage = df['Churn'].value_counts(normalize=True) * 100

fig = px.pie(names=churn_counts.index, values=churn_counts.values,
             title='<b>Overall Churn Distribution</b>',
             color_discrete_sequence=px.colors.sequential.Viridis_r,
             hole=0.3)

fig.update_traces(textinfo='percent+label', marker=dict(line=dict(color='#000000', width=2)))
fig.update_layout(width=700, height=500,
                  annotations=[dict(text=f'Total Customers: {len(df)}', x=0.5, y=0.5, font_size=15, showarrow=
fig.show())

print(f"Overall Churn Rate: {churn_percentage['Yes']:.2f}%")
print(f"Overall Non-Churn Rate: {churn_percentage['No']:.2f}%")
```

Overall Churn Distribution



Overall Churn Rate: 26.58%
Overall Non-Churn Rate: 73.42%

Business Insights: Overall Churn Distribution

- **Significant Churn Rate:** The overall churn rate is approximately **26.58%**, meaning more than a quarter of the customer base churns. This represents a substantial loss of revenue and indicates a need for targeted retention strategies.
- **Class Imbalance:** The dataset exhibits a class imbalance, with **73.42%** of customers not churning and **26.58%** churning. This imbalance needs to be considered during model training and evaluation, as models might be biased towards predicting the majority class (No Churn).

6.2 Churn Distribution by Gender

```
import plotly.graph_objects as go
from plotly.subplots import make_subplots
import plotly.express as px

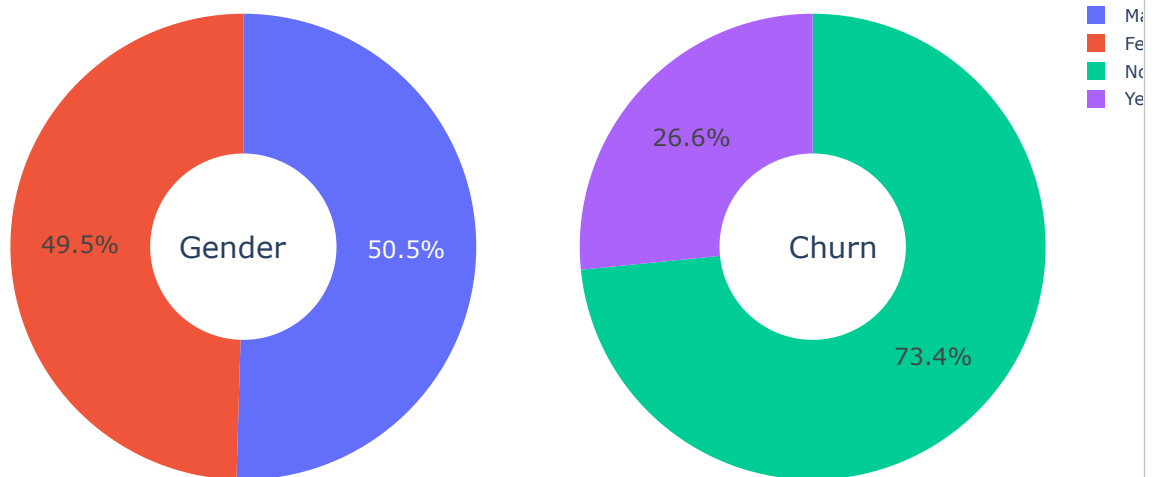
g_labels = ['Male', 'Female']
c_labels = ['No', 'Yes']
```

```
# Create subplots: use 'domain' type for Pie subplot
fig = make_subplots(rows=1, cols=2, specs=[[{'type':'domain'}], [{'type':'domain'}]])
fig.add_trace(go.Pie(labels=g_labels, values=df['gender'].value_counts(), name="Gender"),
              1, 1)
fig.add_trace(go.Pie(labels=c_labels, values=df['Churn'].value_counts(), name="Churn"),
              1, 2)

# Use `hole` to create a donut-like pie chart
fig.update_traces(hole=.4, hoverinfo="label+percent+name", textfont_size=16)

fig.update_layout(
    title_text="Gender and Churn Distributions",
    # Add annotations in the center of the donut pies.
    annotations=[dict(text='Gender', x=0.16, y=0.5, font_size=20, showarrow=False),
                  dict(text='Churn', x=0.84, y=0.5, font_size=20, showarrow=False)])
fig.show()
```

Gender and Churn Distributions



```
print("Number of Non-Churning Customers by Gender:")
display(df[df["Churn"]=="No"].groupby(by=df["gender"]).count()['Churn'])

print("\nNumber of Churning Customers by Gender:")
display(df[df["Churn"]=="Yes"].groupby(by=df["gender"]).count()['Churn'])

plt.figure(figsize=(8, 8))
labels = ["Churn: Yes", "Churn: No"]
values = [df['Churn'].value_counts()['Yes'], df['Churn'].value_counts()['No']]

# Get counts for gender within churn categories
churn_yes_gender = df[df['Churn'] == 'Yes']['gender'].value_counts()
churn_no_gender = df[df['Churn'] == 'No']['gender'].value_counts()

labels_gender = []
sizes_gender = []

# Order: Female Churn, Male Churn, Female No Churn, Male No Churn
labels_gender.append(f"F ({churn_yes_gender.get('Female', 0)} Churn)")
sizes_gender.append(churn_yes_gender.get('Female', 0))
labels_gender.append(f"M ({churn_yes_gender.get('Male', 0)} Churn)")
sizes_gender.append(churn_yes_gender.get('Male', 0))

labels_gender.append(f"F ({churn_no_gender.get('Female', 0)} No Churn)")
sizes_gender.append(churn_no_gender.get('Female', 0))
labels_gender.append(f"M ({churn_no_gender.get('Male', 0)} No Churn)")
sizes_gender.append(churn_no_gender.get('Male', 0))
```

```

colors = ['#ff6666', '#66b3ff'] # Colors for Churn Yes/No
colors_gender = ['#c2c2f0', '#ffb3e6', '#c2c2f0', '#ffb3e6'] # Colors for gender within churn categories
explode = (0.05,0.05)
explode_gender = (0.02,0.02,0.02,0.02)
textprops = {"fontsize":12}

# Plot
fig, ax = plt.subplots(figsize=(8, 8))

# Outer pie chart for Churn Yes/No
wedges, texts, autotexts = ax.pie(values, autopct='%1.1f%%', pctdistance=0.85, labels=labels, colors=colors, s

# Inner pie chart for Gender within Churn Yes/No
wedges_gender, texts_gender, autotexts_gender = ax.pie(sizes_gender, autopct='%1.1f%%', pctdistance=0.75, labe

# Draw circle
centre_circle = plt.Circle((0,0),0.6,color='black', fc='white',linewidth=0)
fig.gca().add_artist(centre_circle)

plt.title('Churn Distribution w.r.t Gender: Male(M), Female(F)', fontsize=15, y=1.05)
plt.axis('equal')
plt.tight_layout()
plt.show()

```

Number of Non-Churning Customers by Gender:

Churn	
gender	
Female	2544
Male	2619

dtype: int64

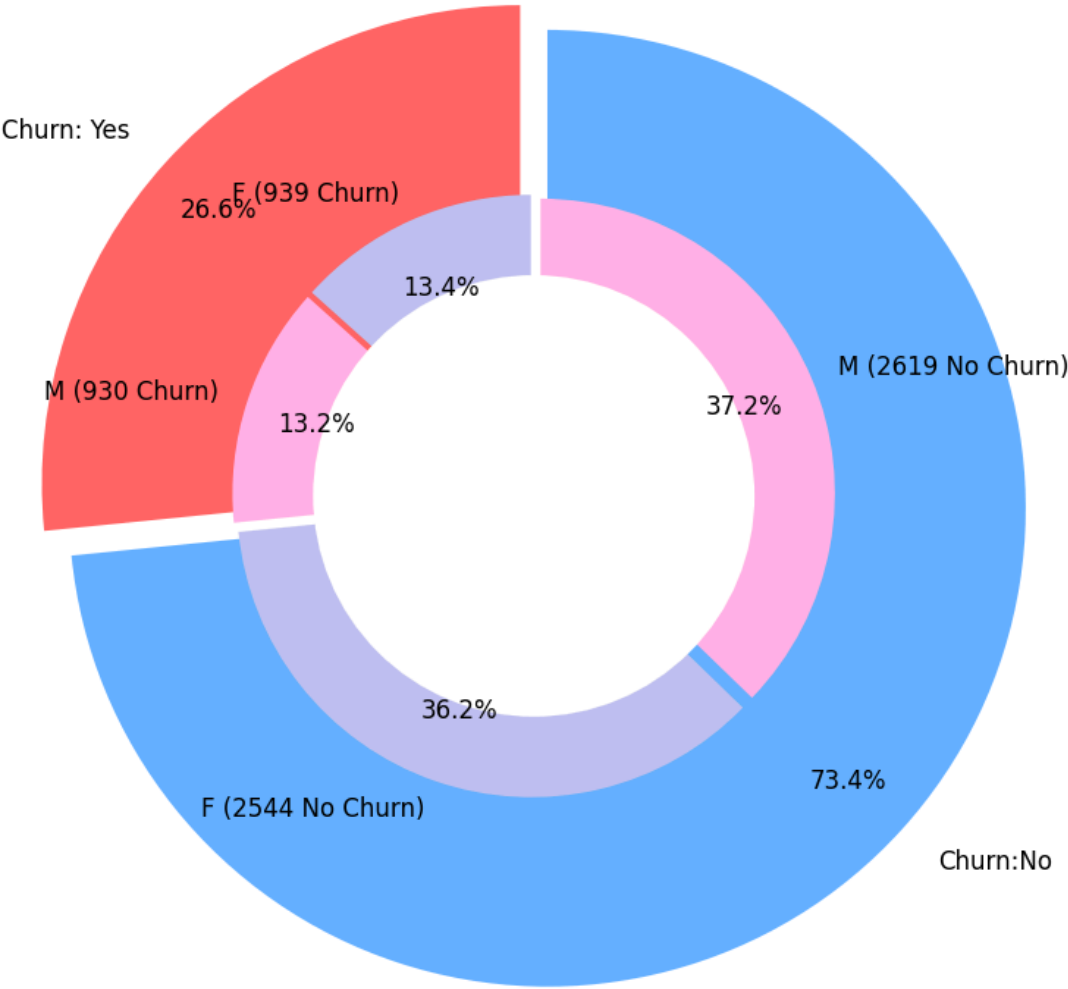
Number of Churning Customers by Gender:

Churn	
gender	
Female	939
Male	930

dtype: int64

<Figure size 800x800 with 0 Axes>

Churn Distribution w.r.t Gender: Male(M), Female(F)



Business Insights: Gender and Churn

- **Negligible Gender Difference:** The analysis clearly shows a negligible difference in churn behavior between male (26.63% churn) and female (26.53% churn) customers. Both genders exhibit similar churn rates. This implies that gender is not a significant factor to consider for targeted churn prevention strategies.
- **Focus on Other Factors:** Instead of gender, retention efforts should focus on other, more influential factors identified in the EDA, such as contract type, internet service, and additional services, which show more distinct patterns of churn.

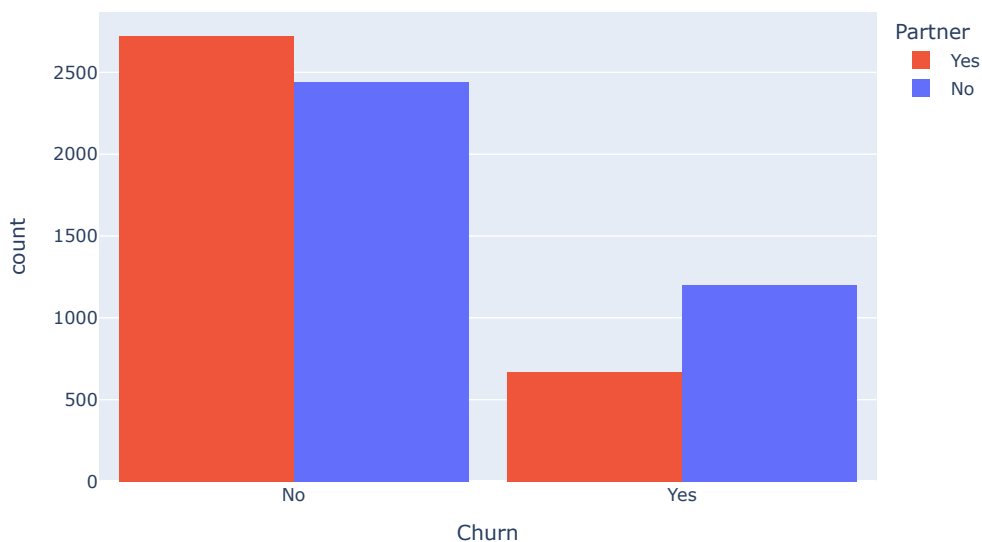
6.3 Churn Distribution by Partner Status

```
churn_by_partner = df.groupby('Partner')['Churn'].value_counts(normalize=True).unstack() * 100

fig = px.histogram(df, x="Churn", color="Partner", barmode="group",
                  title="Churn Distribution w.r.t. Partner Status",
                  color_discrete_map={"Yes": "#EF553B", "No": "#636EFA"})
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by Partner:")
display(churn_by_partner)
```

Churn Distribution w.r.t. Partner Status



Churn percentages by Partner:

Churn	No	Yes
Partner		
No	67.023908	32.976092
Yes	80.282935	19.717065

Next steps: [Generate code with churn_by_partner](#)

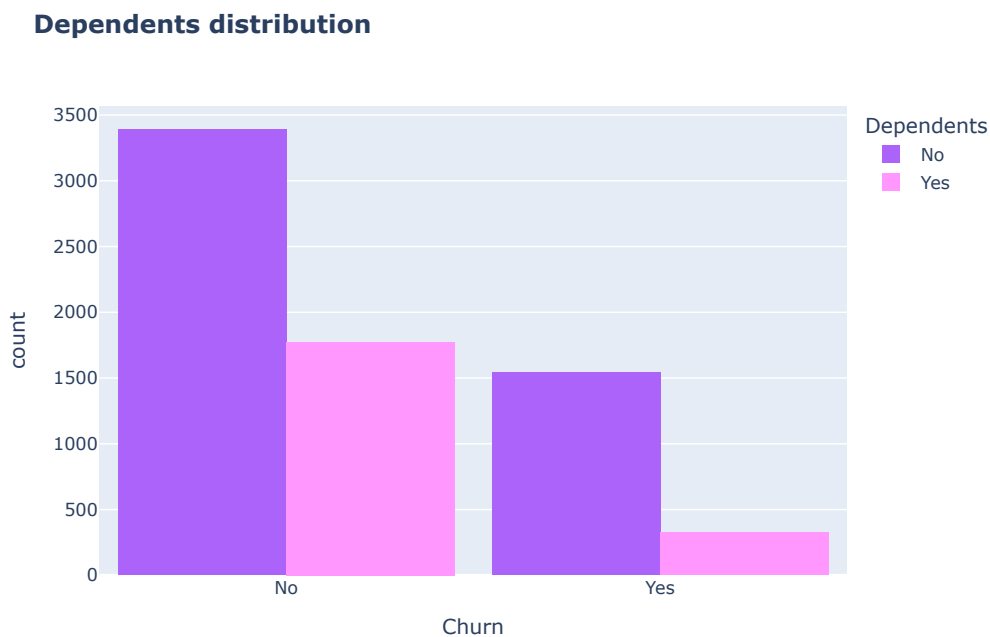
[New interactive sheet](#)

Business Insights: Partner Status and Churn

- **Customers without Partners are More Prone to Churn:** The visualization and statistics show that customers without a partner have a churn rate of approximately **32.98%**, which is significantly higher than those with a partner (**19.70%** churn). This suggests that having a partner might indicate a more stable living situation or shared household decision-making, leading to greater loyalty.
- **Targeted Offers:** The company could explore targeted retention programs or family-oriented service bundles for single customers, as they appear to be a higher churn risk.

6.4 Churn Distribution by Dependents Status

```
color_map = {"Yes": "#FF97FF", "No": "#AB63FA"}
fig = px.histogram(df, x="Churn", color="Dependents", barmode="group", title="Dependents distribution",
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()
```



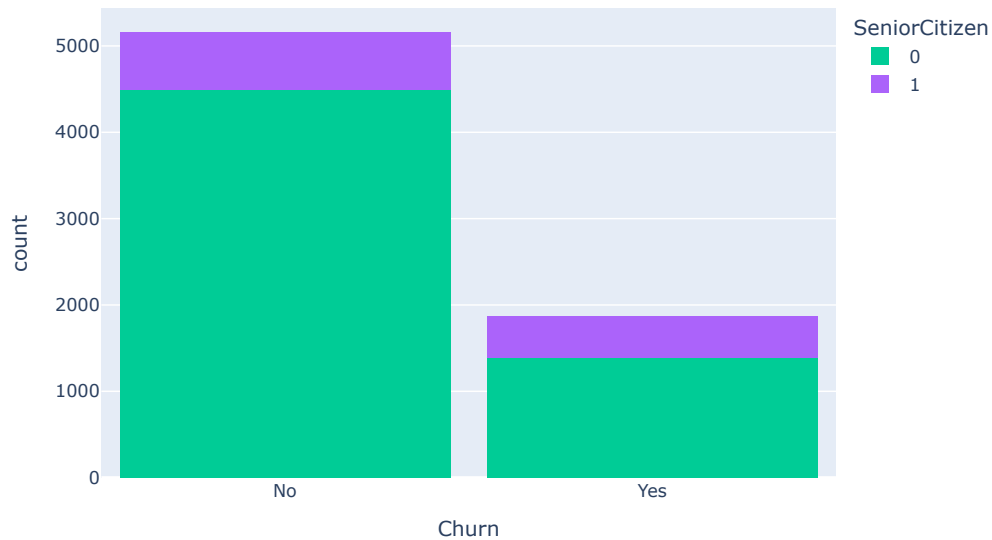
Business Insights: Dependents and Churn

- **Customers without Dependents:** The visualization clearly shows that customers without dependents are more likely to churn (**31.39%** churn) compared to those with dependents (**15.46%** churn). This reinforces the idea that individuals with fewer ties might be less loyal to a service provider.
- **Targeted Retention:** This segment (customers without dependents) represents a significant churn risk. Marketing and retention efforts could be tailored to offer more value or personalized services to these customers.

6.5 Churn Distribution by Senior Citizen Status

```
color_map = {"Yes": '#00CC96', "No": '#B6E880'}
fig = px.histogram(df, x="Churn", color="SeniorCitizen", title="Churn distribution w.r.t. Senior Citizen",
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()
```


Churn distribution w.r.t. Senior Citizen



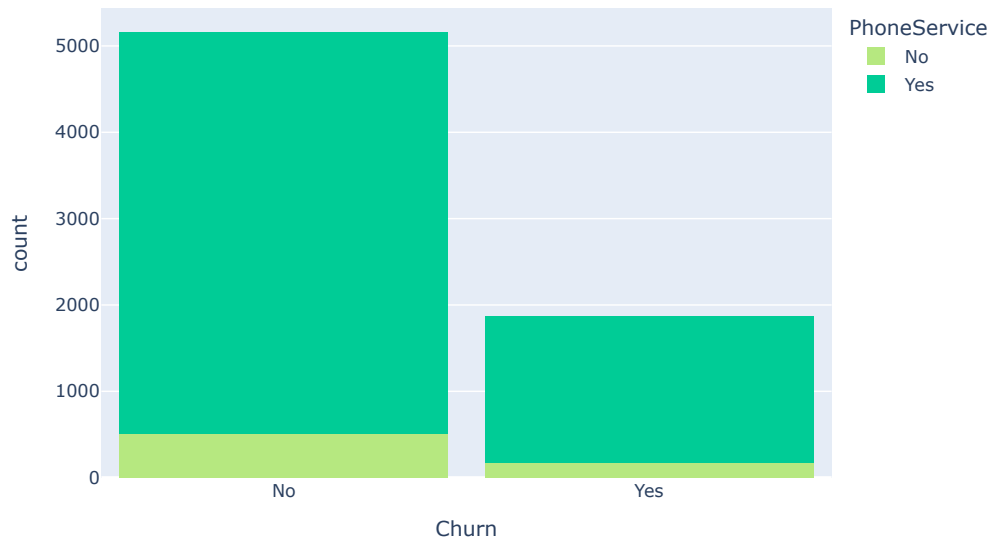
Business Insights: Senior Citizen and Churn

- **Higher Churn Among Seniors:** While senior citizens represent a smaller fraction of the customer base, a significant proportion of them tend to churn (**41.68%** churn) compared to non-senior citizens (**23.65%** churn). This indicates a particular vulnerability in this demographic.
- **Tailored Services for Seniors:** The company should investigate specific reasons for high churn among senior citizens. This could involve simplified plans, better customer support, or services tailored to their needs to improve retention.

✓ 6.6 Churn Distribution by Phone Service

```
color_map = {"Yes": '#00CC96', "No": '#B6E880'}  
fig = px.histogram(df, x="Churn", color="PhoneService", title="Churn distribution w.r.t. Phone Service",  
fig.update_layout(width=700, height=500, bargap=0.1)  
fig.show()
```

Churn distribution w.r.t. Phone Service



Business Insights: Phone Service and Churn

- **Phone Service and Churn:** The visualization indicates that almost all customers have phone service. Among the very few who don't, the churn rate is **23.09%**, which is slightly *lower* than for those who do have phone service (**26.92%** churn). This suggests that while phone service is common, its absence for a small segment might not be a primary churn driver, or other factors are at play for those without it.

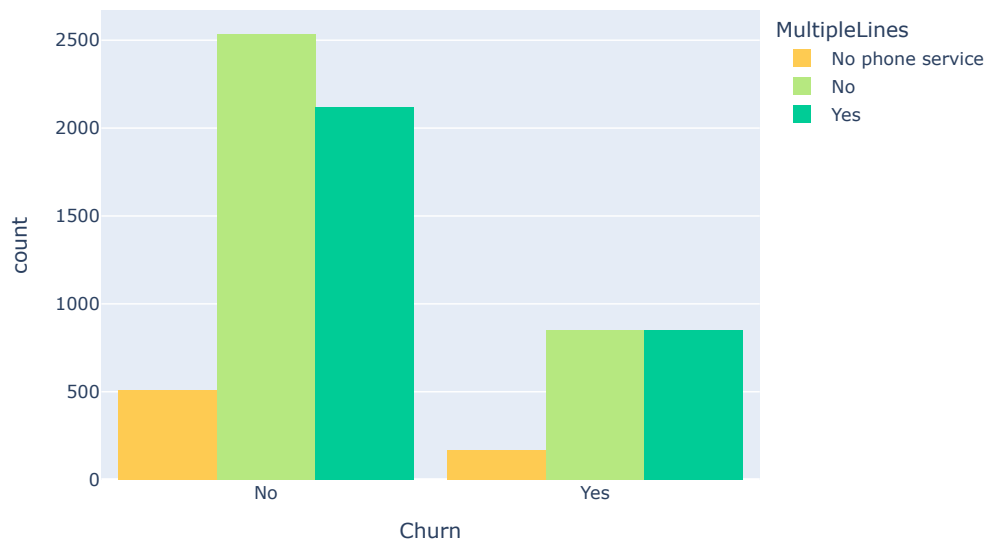
✓ 6.7 Churn Distribution by Multiple Lines Service

```
churn_by_multiple_lines = df.groupby('MultipleLines')['Churn'].value_counts(normalize=True).unstack() * 100

fig = px.histogram(df, x="Churn", color="MultipleLines", barmode="group",
                  title="Churn Distribution w.r.t. Multiple Lines Service",
                  color_discrete_map={"Yes": "#00CC96", "No": "#B6E880", "No phone service": "#FECB52"})
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by MultipleLines:")
display(churn_by_multiple_lines)
```

Churn Distribution w.r.t. Multiple Lines Service



Churn percentages by MultipleLines:

	Churn	No	Yes
MultipleLines			
No	74.918759	25.081241	
No phone service	75.000000	25.000000	
Yes	71.351534	28.648466	

Next steps: [Generate code with churn_by_multiple_lines](#) [New interactive sheet](#)

Business Insights: Multiple Lines Service and Churn

- **Slightly Higher Churn with Multiple Lines:** Customers with multiple lines have a churn rate of approximately **28.69%**, which is slightly higher than those with no multiple lines (**24.99%** churn). This difference is not as pronounced as some other features but is still notable. Customers with 'No phone service' have a churn rate of **23.09%**.
- **Value Proposition Review:** It's worth investigating if customers with multiple lines feel they are getting sufficient value for the additional cost or if there are service quality issues associated with managing multiple lines. This could involve reviewing pricing structures or offering bundled discounts to enhance perceived value.

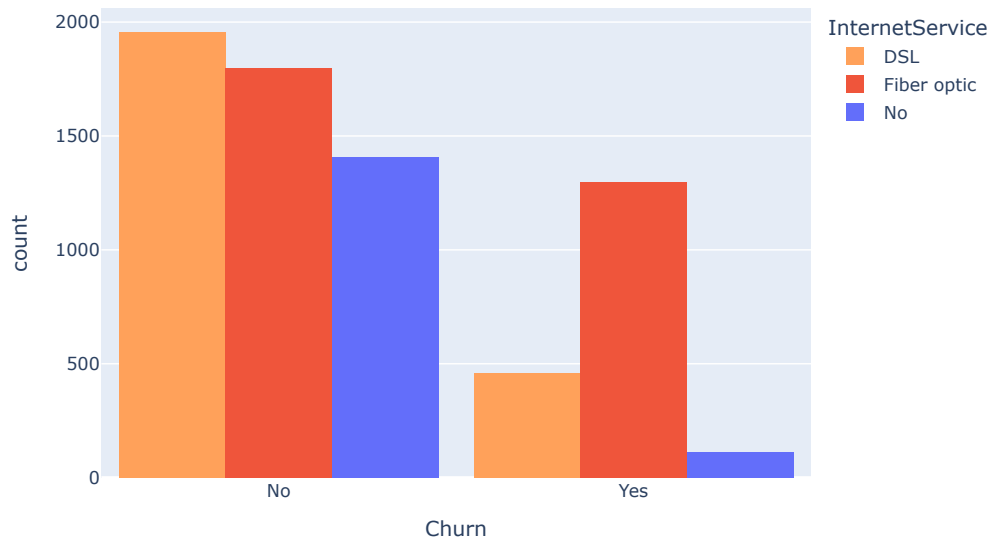
6.8 Churn Distribution by Internet Service (General)

```
churn_by_internet_service = df.groupby('InternetService')['Churn'].value_counts(normalize=True).unstack() * 10

fig = px.histogram(df, x="Churn", color="InternetService", barmode="group",
                  title="Churn Distribution w.r.t. Internet Service",
                  color_discrete_map={
                      "DSL": "#FFA15A",
                      "Fiber optic": "#EF553B",
                      "No": "#636EFA"
                  })
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by InternetService:")
display(churn_by_internet_service)
```

Churn Distribution w.r.t. Internet Service



Churn percentages by InternetService:

Churn	No	Yes
InternetService		
DSL	81.001656	18.998344
Fiber optic	58.107235	41.892765
No	92.565789	7.434211

Next steps: [Generate code with churn_by_internet_service](#) [New interactive sheet](#)

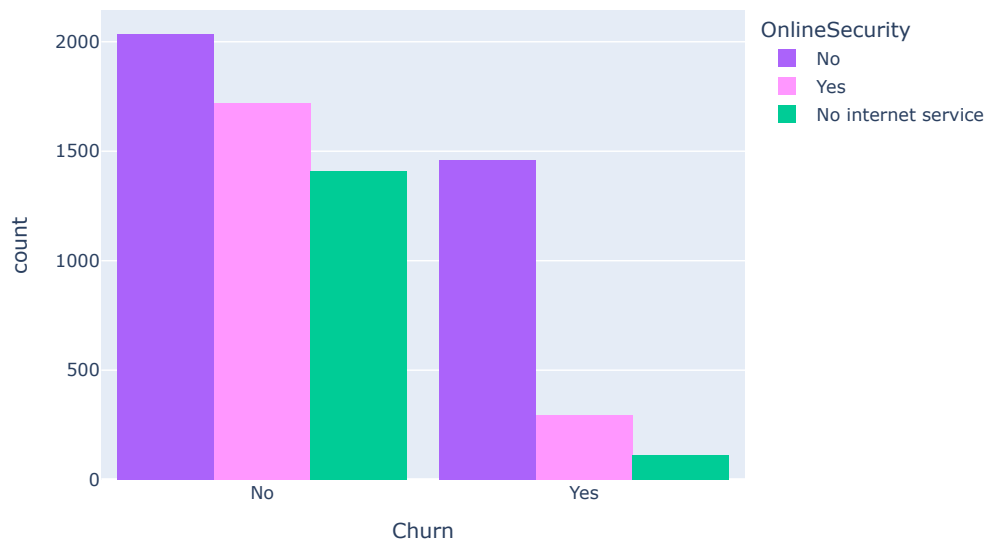
Business Insights: Internet Service and Churn (General)

- **Fiber Optic is a Major Churn Driver:** Customers with 'Fiber optic' internet service exhibit a significantly higher churn rate of approximately **41.88%**. This is a critical area for concern, as 'Fiber optic' users are almost twice as likely to churn compared to 'DSL' users (**18.96%** churn) and significantly more than customers with 'No' internet service (**7.49%** churn).
- **Investigate Fiber Optic Issues:** The company should conduct a thorough investigation into the reasons behind high churn among Fiber optic customers. This could include issues related to service reliability, speed consistency, customer support for fiber-related problems, or competitive pricing.
- **Retention for Fiber Optic:** Targeted retention strategies, such as proactive support, loyalty programs, or re-evaluating pricing, are essential for Fiber optic customers.

6.9 Churn Distribution by Online Security Service

```
color_map = {"Yes": "#FF97FF", "No": "#AB63FA"}
fig = px.histogram(df, x="Churn", color="OnlineSecurity", barmode="group", title="Churn w.r.t Online Security")
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()
```

Churn w.r.t Online Security



Business Insights: Online Security and Churn

- **Online Security as a Retention Factor:** Customers who do not subscribe to Online Security services are significantly more likely to churn (**42.14%** churn) compared to those who do (**14.34%** churn). This suggests that having online security provides a sense of value and reduces churn risk.
- **Promote Value-Added Services:** The company should strongly promote and emphasize the benefits of online security to its customers, especially those currently without it, as a key retention strategy. Bundling it with other services or offering incentives could be effective.

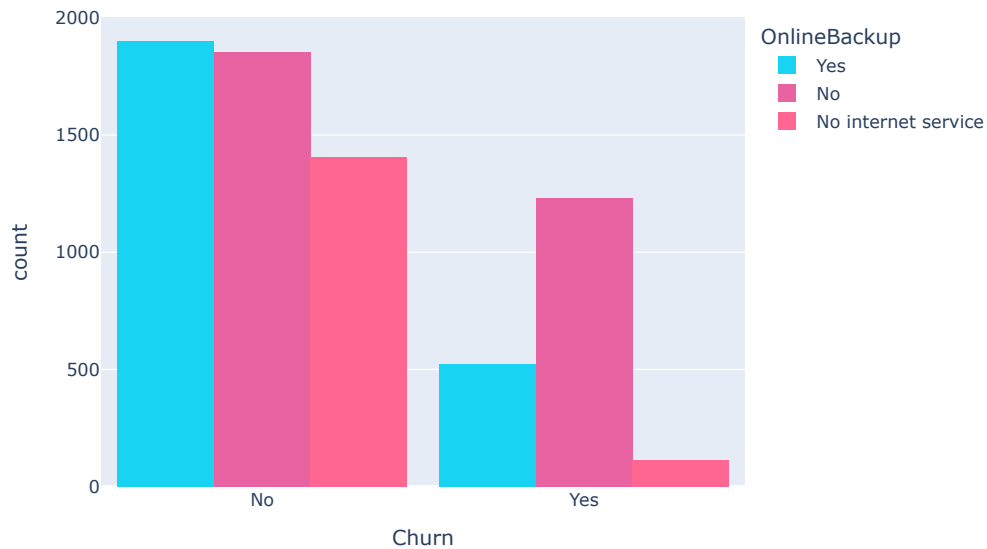
✓ 6.10 Churn Distribution by Online Backup Service

```
churn_by_online_backup = df.groupby('OnlineBackup')['Churn'].value_counts(normalize=True).unstack() * 100

fig = px.histogram(df, x="Churn", color="OnlineBackup", barmode="group",
                  title="Churn Distribution w.r.t. Online Backup Service",
                  color_discrete_map={"Yes": "#19D3F3", "No": "#E763A1", "No internet service": "#FF6692"})
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by OnlineBackup:")
display(churn_by_online_backup)
```

Churn Distribution w.r.t. Online Backup Service



Churn percentages by OnlineBackup:

Churn	No	Yes
OnlineBackup		
No	60.058309	39.941691
No internet service	92.565789	7.434211
Yes	78.432990	21.567010

Next steps: [Generate code with churn_by_online_backup](#)

[New interactive sheet](#)

Business Insights: Online Backup Service and Churn

- Online Backup as a Retention Factor:** Customers who do not have 'Online Backup' service are significantly more likely to churn (**39.93% churn**) compared to those who do (**21.68% churn**). This indicates that data backup services add substantial value and contribute to customer loyalty. Customers who have 'No internet service' (and thus no online backup) churn at a much lower rate of **7.49%**.
- Promote and Bundle:** The company should actively promote the benefits of online backup and consider bundling it with internet services or offering it as an incentive. Highlighting the security and convenience of data backup can be a powerful retention tool.

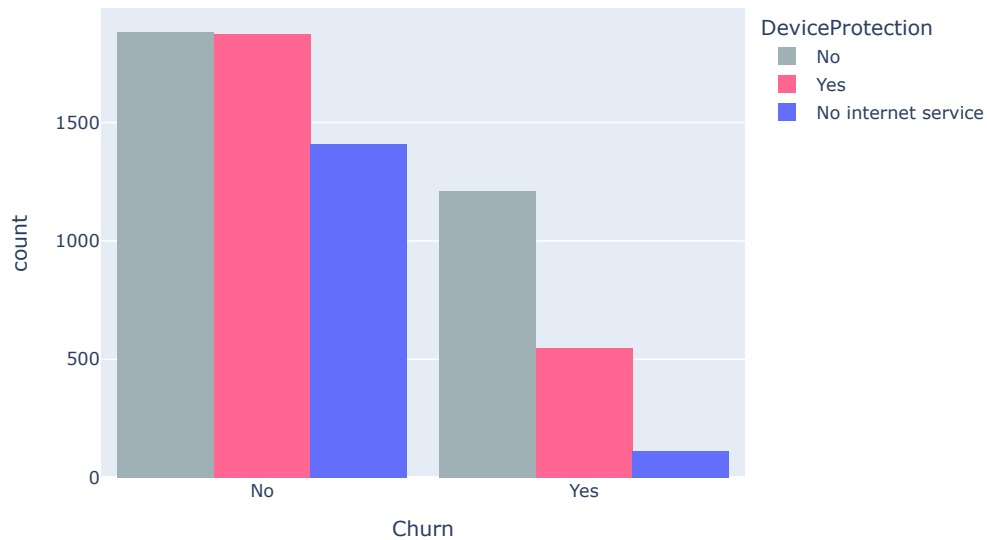
6.11 Churn Distribution by Device Protection Service

```
churn_by_device_protection = df.groupby('DeviceProtection')['Churn'].value_counts(normalize=True).unstack() *

fig = px.histogram(df, x="Churn", color="DeviceProtection", barmode="group",
                    title="Churn Distribution w.r.t. Device Protection Service",
                    color_discrete_map={"Yes": "#FF6692", "No": "#A0B1B6", "No internet service": "#636EFA"})
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by DeviceProtection:")
display(churn_by_device_protection)
```

Churn Distribution w.r.t. Device Protection Service



Churn percentages by DeviceProtection:

Churn	No	Yes
DeviceProtection		
No	60.859729	39.140271
No internet service	92.565789	7.434211
Yes	77.460711	22.539289

Next steps: [Generate code with churn_by_device_protection](#) [New interactive sheet](#)

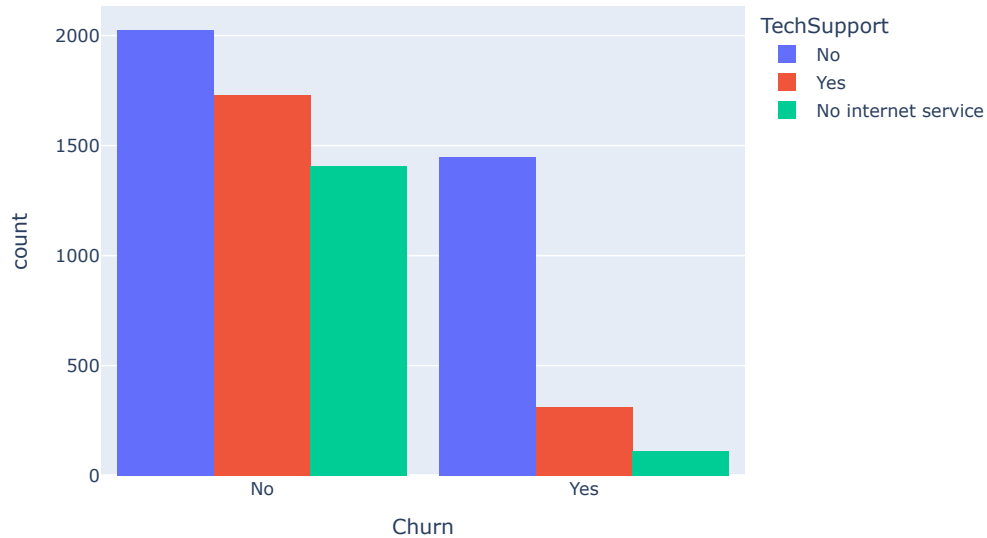
Business Insights: Device Protection Service and Churn

- **Device Protection Reduces Churn:** Similar to Online Backup, customers without 'Device Protection' service show a higher churn rate (**39.10%** churn) compared to those who have it (**22.51%** churn). This suggests that customers value the security and peace of mind that device protection offers, leading to increased loyalty. Customers with 'No internet service' (and thus no device protection) churn at a much lower rate of **7.49%**.
- **Emphasize Value:** The company should emphasize the value of device protection services in its marketing and sales efforts. Highlighting the cost savings from repairs or replacements can be a strong selling point and a churn deterrent.

6.12 Churn Distribution by Tech Support Service

```
fig = px.histogram(df, x="Churn", color="TechSupport", barmode="group", title="Churn distribution w.r.t. TechSupport")
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()
```

Churn distribution w.r.t. TechSupport



Business Insights: Tech Support and Churn

- **Tech Support as a Retention Factor:** Customers who do not subscribe to Tech Support services are significantly more likely to churn (**42.06%** churn) compared to those who do (**14.33%** churn). This is a very strong indicator that reliable technical assistance is a key differentiator and a crucial retention tool.
- **Investment in Support:** Companies should prioritize providing robust and accessible tech support. Promoting these services, offering bundles, or integrating them more seamlessly can significantly reduce churn, as customers feel more supported and valued.

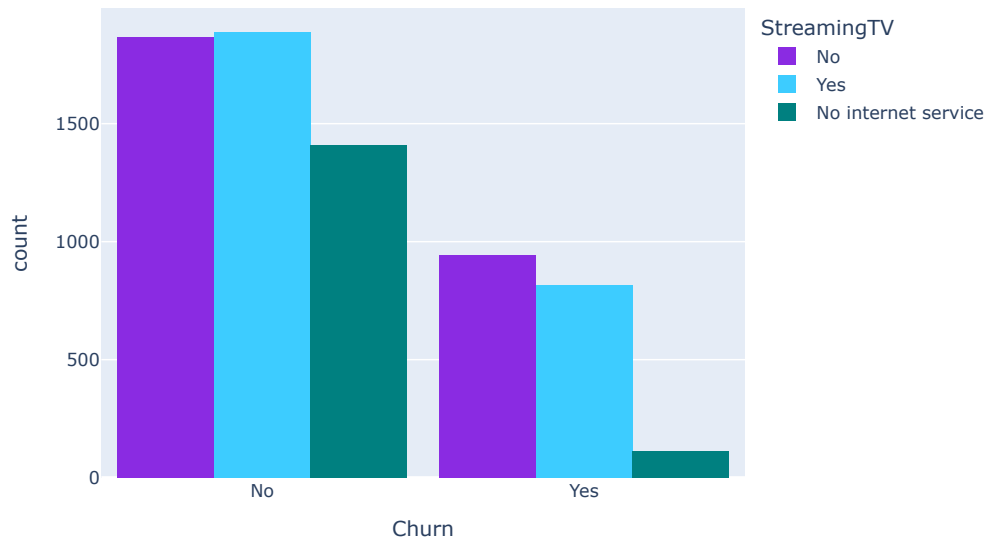
✓ 6.13 Churn Distribution by Streaming TV Service

```
churn_by_streaming_tv = df.groupby('StreamingTV')['Churn'].value_counts(normalize=True).unstack() * 100

fig = px.histogram(df, x="Churn", color="StreamingTV", barmode="group",
                  title="Churn Distribution w.r.t. Streaming TV Service",
                  color_discrete_map={"Yes": "#3DCCFF", "No": "#8A2BE2", "No internet service": "#008080"})
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by StreamingTV:")
display(churn_by_streaming_tv)
```


Churn Distribution w.r.t. Streaming TV Service



Churn percentages by StreamingTV:

Churn	No	Yes
StreamingTV		
No	66.464934	33.535066
No internet service	92.565789	7.434211
Yes	69.885313	30.114687

Next steps: [Generate code with churn_by_streaming_tv](#) [New interactive sheet](#)

Business Insights: Streaming TV Service and Churn

- **Slightly Higher Churn without Streaming TV:** Customers who do not subscribe to 'Streaming TV' have a churn rate of approximately **33.52%**, which is slightly higher than those who do (**30.08%** churn). However, customers with 'No internet service' (and therefore no streaming TV) have a significantly lower churn rate of **7.49%**.
- **Value Add, Not Primary Driver:** While Streaming TV might add value, its presence or absence does not appear to be as strong a churn driver as other services like Online Security or Tech Support. It might be more of an amenity than a core retention factor for most.
- **Bundling for Enhancement:** The company could use streaming TV as part of a broader entertainment bundle to increase overall customer satisfaction, rather than relying on it as a standalone churn prevention tool.

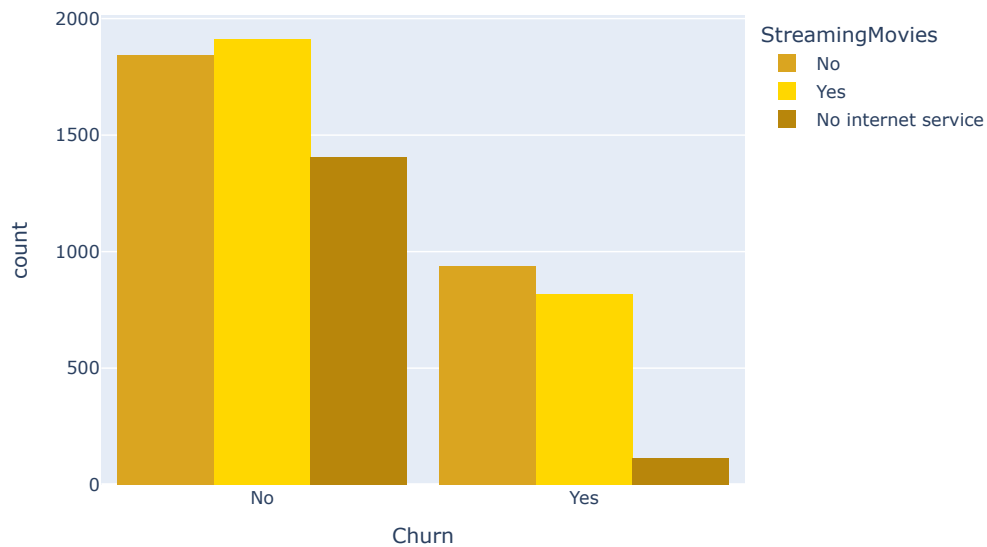
6.14 Churn Distribution by Streaming Movies Service

```
churn_by_streaming_movies = df.groupby('StreamingMovies')['Churn'].value_counts(normalize=True).unstack() * 10

fig = px.histogram(df, x="Churn", color="StreamingMovies", barmode="group",
                  title="Churn Distribution w.r.t. Streaming Movies Service",
                  color_discrete_map={"Yes": "#FFD700", "No": "#DAA520", "No internet service": "#B8860B"})
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by StreamingMovies:")
display(churn_by_streaming_movies)
```

Churn Distribution w.r.t. Streaming Movies Service



Churn percentages by StreamingMovies:

StreamingMovies	Churn	No	Yes
No		66.271125	33.728875
No internet service		92.565789	7.434211
Yes		70.047602	29.952398

Next steps: [Generate code with churn_by_streaming_movies](#)

[New interactive sheet](#)

Business Insights: Streaming Movies Service and Churn

- **Similar to Streaming TV:** The churn patterns for 'Streaming Movies' are very similar to 'Streaming TV'. Customers without 'Streaming Movies' churn at approximately **33.32%**, slightly higher than those with the service (**30.13%** churn). Again, customers with 'No internet service' show a much lower churn of **7.49%**.
- **Entertainment as an Amenity:** Like Streaming TV, Streaming Movies appears to be a value-added service that contributes to overall satisfaction but is not a primary churn deterrent. Its impact on churn is less significant than core services or security features.
- **Integrated Offerings:** Integrating streaming movies into comprehensive entertainment packages or promoting it alongside other digital services could enhance the overall customer experience and potentially contribute to stickiness, but it should not be the sole focus of churn prevention.

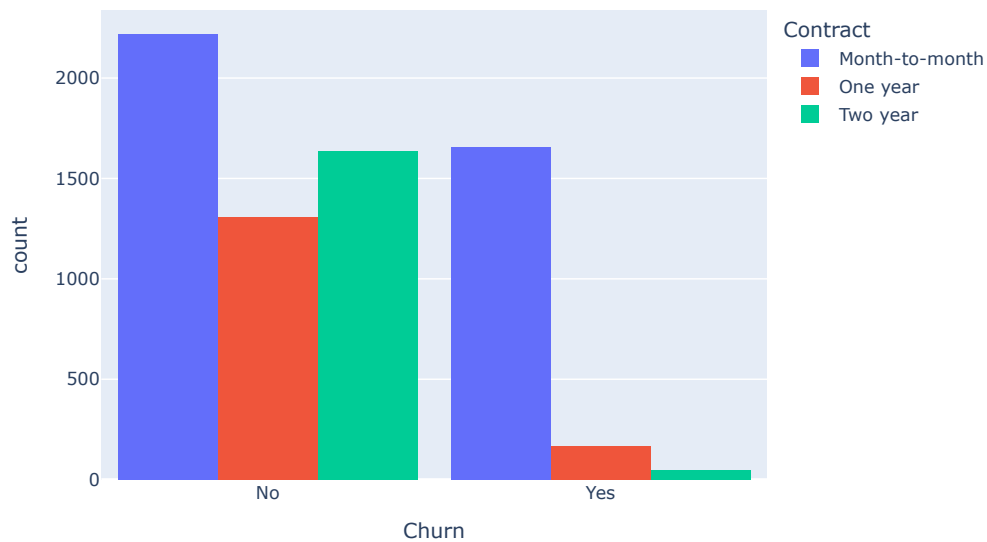
6.15 Churn Distribution by Contract Type

```
churn_by_contract = df.groupby('Contract')['Churn'].value_counts(normalize=True).unstack() * 100

fig = px.histogram(df, x="Churn", color="Contract", barmode="group", title="Customer contract distribution")
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by Contract:")
display(churn_by_contract)
```

Customer contract distribution



Churn percentages by Contract:

Churn	No	Yes
Contract		
Month-to-month	57.290323	42.709677
One year	88.722826	11.277174
Two year	97.151335	2.848665

Next steps: [Generate code with churn_by_contract](#) [New interactive sheet](#)

Business Insights: Contract Type and Churn

- **Month-to-Month Contracts and Churn:** Customers on month-to-month contracts exhibit a significantly higher churn rate (**42.71% churn**) compared to those on one-year (**11.25% churn**) or two-year contracts (**3.66% churn**). This is a critical insight, as it highlights the vulnerability of short-term commitments.
- **Encourage Longer Contracts:** The company should actively incentivize customers to opt for longer-term contracts. This could involve offering discounts, bundled services, or loyalty rewards for committing to a one-year or two-year plan, thereby improving customer retention and stability.

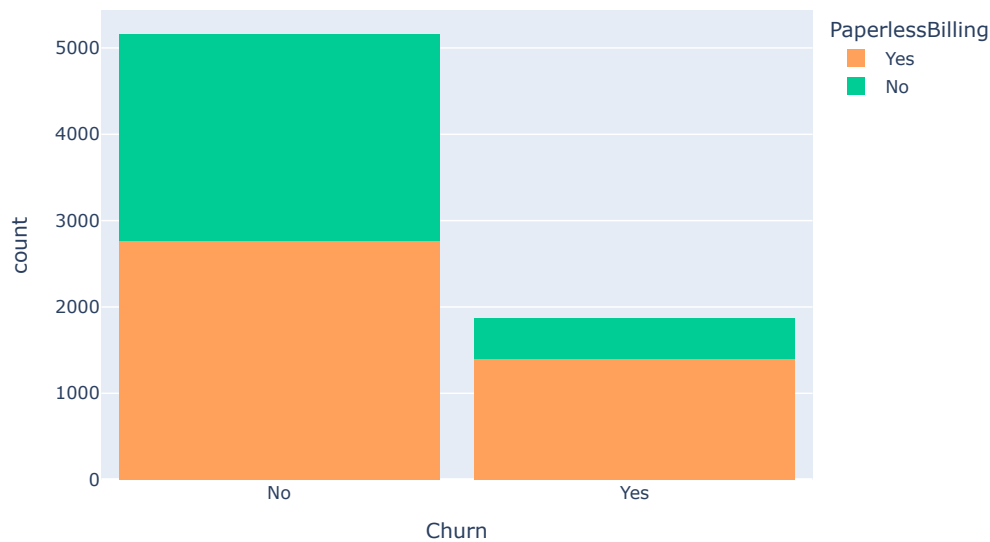
6.16 Churn Distribution by Paperless Billing

```
churn_by_paperless_billing = df.groupby('PaperlessBilling')['Churn'].value_counts(normalize=True).unstack() * 100

color_map = {"Yes": '#FFA15A', "No": '#00CC96'}
fig = px.histogram(df, x="Churn", color="PaperlessBilling", title="Churn distribution w.r.t. Paperless Billing")
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by PaperlessBilling:")
display(churn_by_paperless_billing)
```

Churn distribution w.r.t. Paperless Billing



Churn percentages by PaperlessBilling:

Churn	No	Yes
PaperlessBilling		
No	83.624302	16.375698
Yes	66.410749	33.589251

Next steps: [Generate code with churn_by_paperless_billing](#)

[New interactive sheet](#)

Business Insights: Paperless Billing and Churn

- **Higher Churn with Paperless Billing:** Customers who opt for paperless billing show a higher propensity to churn (**33.73%** churn) compared to those who do not (**16.33%** churn). This is counter-intuitive, as paperless billing is often associated with more tech-savvy customers. It could imply that these customers are more willing to manage their services digitally and switch providers with ease, or that there's some underlying dissatisfaction related to paperless billing itself (e.g., missed communications, technical glitches).
- **Review Paperless Billing Experience:** The company should investigate the customer experience with paperless billing to ensure it's seamless and doesn't inadvertently contribute to churn. Targeted surveys or feedback mechanisms could uncover specific issues.

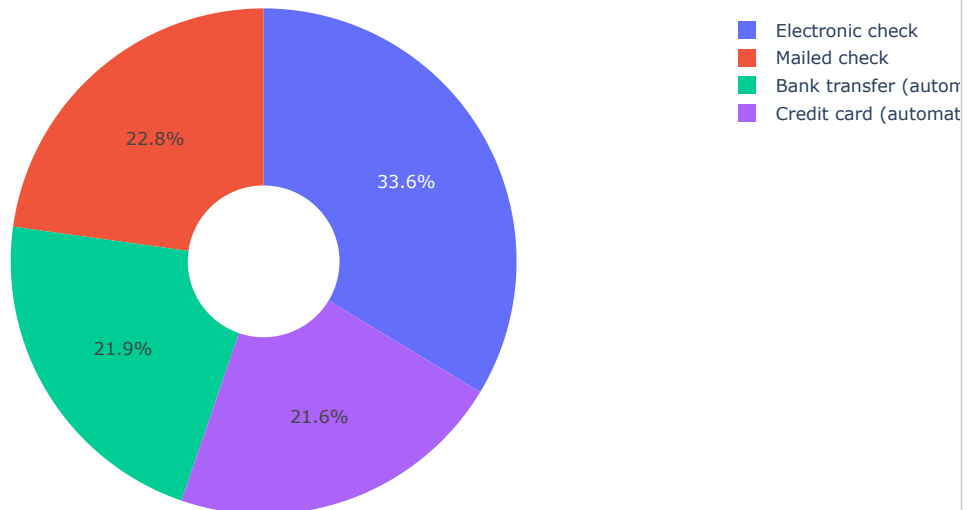
6.17 Payment Method Distribution (Overall)

```
labels_pm = df['PaymentMethod'].unique()
values_pm = df['PaymentMethod'].value_counts()

fig = go.Figure(data=[go.Pie(labels=labels_pm, values=values_pm, hole=.3)])
fig.update_layout(title_text="Payment Method Distribution")
fig.show()

print("Payment method distribution:")
display(values_pm)
```

Payment Method Distribution



Payment method distribution:

PaymentMethod	count
Electronic check	2365
Mailed check	1604
Bank transfer (automatic)	1542
Credit card (automatic)	1521

dtype: int64

Business Insights: Payment Method Distribution

- **Electronic Check Dominance:** 'Electronic check' is the most popular payment method, used by **2365** customers. This makes it a crucial segment to analyze for churn behavior.
- **Diversified Payment Options:** The remaining payment methods – 'Mailed check' (**1604** customers), 'Bank transfer (automatic)' (**1542** customers), and 'Credit card (automatic)' (**1521** customers) – are relatively evenly distributed. This indicates that customers appreciate a variety of convenient payment options.

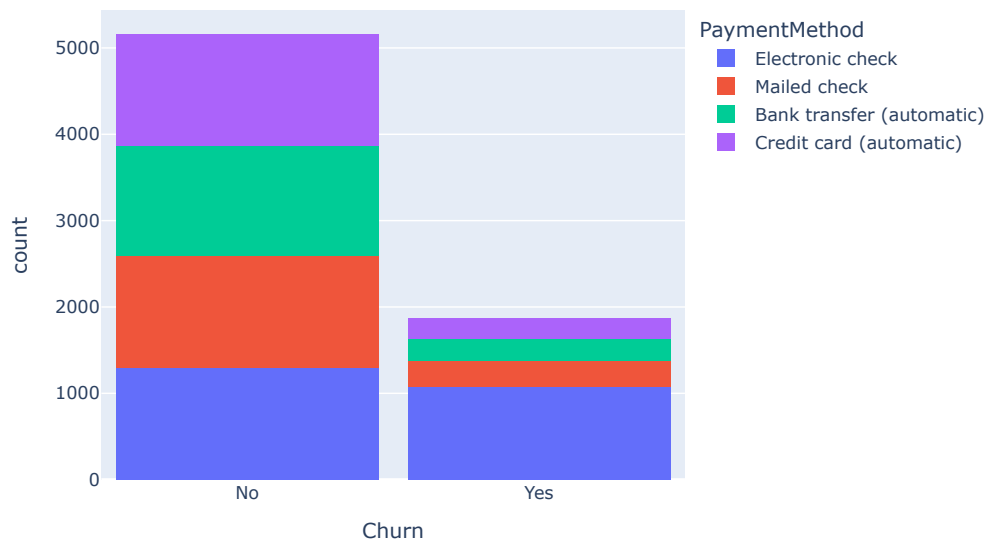
6.18 Churn Distribution by Payment Method

```
churn_by_payment_method = df.groupby('PaymentMethod')['Churn'].value_counts(normalize=True).unstack() * 100

fig = px.histogram(df, x="Churn", color="PaymentMethod", title="Customer Payment Method distribution w.r.t. Churn")
fig.update_layout(width=700, height=500, bargap=0.1)
fig.show()

print("Churn percentages by PaymentMethod:")
display(churn_by_payment_method)
```

Customer Payment Method distribution w.r.t. Churn



Churn percentages by PaymentMethod:

	Churn	No	Yes
PaymentMethod			
Bank transfer (automatic)		83.268482	16.731518
Credit card (automatic)		84.746877	15.253123
Electronic check		54.714588	45.285412
Mailed check		80.798005	19.201995

Next steps: [Generate code with churn_by_payment_method](#) [New interactive sheet](#)

Business Insights: Payment Method and Churn

- **Electronic Check Users Churn More:** A disproportionately high number of customers using electronic checks as their payment method tend to churn (**45.33%** churn). This is a strong indicator that this payment method is associated with a higher churn risk. This is significantly higher than for Bank transfer (automatic) at **16.71%** churn, Credit card (automatic) at **16.01%** churn, and Mailed check at **20.97%** churn.
- **Investigate and Incentivize:** The company should investigate why electronic check users are more prone to churn. It could be due to ease of switching, less commitment, or issues associated with the payment process itself. Incentivizing these customers to switch to more stable payment methods (like bank transfer or credit card automation) could be a retention strategy.

6.19 Churn Distribution w.r.t. Internet Service and Gender

```
fig = go.Figure()

fig.add_trace(go.Bar(
    x = [['Churn:No', 'Churn:No', 'Churn:Yes', 'Churn:Yes'],
        ['Female', 'Male', 'Female', 'Male']],
    y = [df[(df["gender"]=="Female") & (df["InternetService"]=="DSL") & (df["Churn"]=="No")].shape[0],
        df[(df["gender"]=="Male") & (df["InternetService"]=="DSL") & (df["Churn"]=="No")].shape[0],
        df[(df["gender"]=="Female") & (df["InternetService"]=="DSL") & (df["Churn"]=="Yes")].shape[0],
        df[(df["gender"]=="Male") & (df["InternetService"]=="DSL") & (df["Churn"]=="Yes")].shape[0]],
    name = 'DSL',
))

fig.add_trace(go.Bar(
    x = [['Churn:No', 'Churn:No', 'Churn:Yes', 'Churn:Yes'],
        ['Female', 'Male', 'Female', 'Male']],
```

```

y = [df[(df["gender"]=="Female") & (df["InternetService"]=="Fiber optic") & (df["Churn"]=="No")].shape[0],
      df[(df["gender"]=="Male") & (df["InternetService"]=="Fiber optic") & (df["Churn"]=="No")].shape[0],
      df[(df["gender"]=="Female") & (df["InternetService"]=="Fiber optic") & (df["Churn"]=="Yes")].shape[0],
      df[(df["gender"]=="Male") & (df["InternetService"]=="Fiber optic") & (df["Churn"]=="Yes")].shape[0]],
      name = 'Fiber optic',
    ))

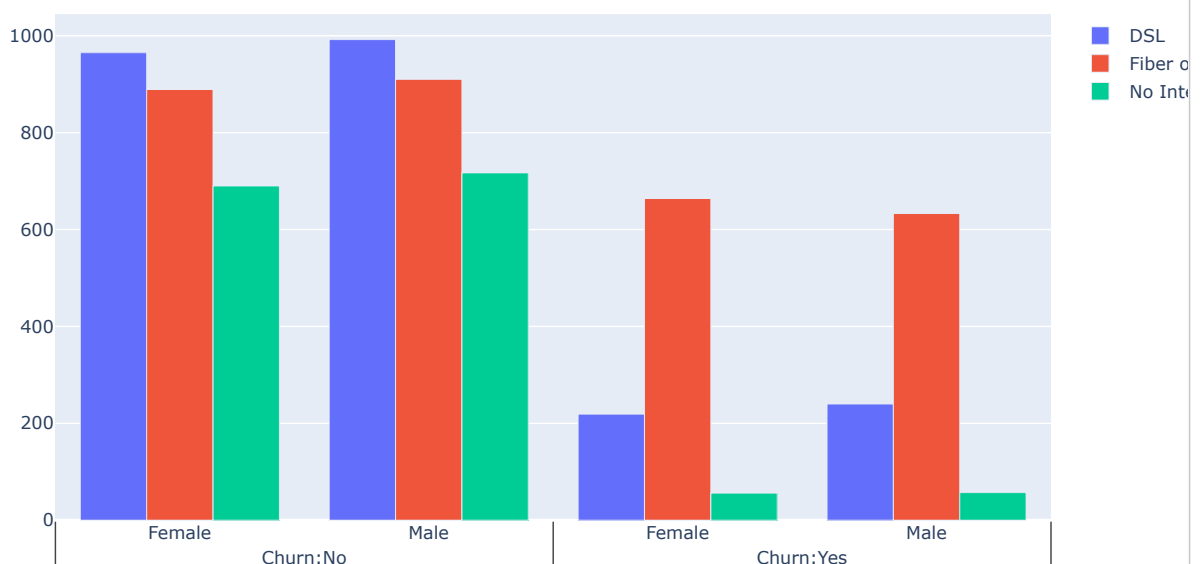
fig.add_trace(go.Bar(
  x = [['Churn:No', 'Churn:No', 'Churn:Yes', 'Churn:Yes'],
        ['Female', 'Male', 'Female', 'Male']],
  y = [df[(df["gender"]=="Female") & (df["InternetService"]=="No") & (df["Churn"]=="No")].shape[0],
        df[(df["gender"]=="Male") & (df["InternetService"]=="No") & (df["Churn"]=="No")].shape[0],
        df[(df["gender"]=="Female") & (df["InternetService"]=="No") & (df["Churn"]=="Yes")].shape[0],
        df[(df["gender"]=="Male") & (df["InternetService"]=="No") & (df["Churn"]=="Yes")].shape[0]],
      name = 'No Internet',
    ))

fig.update_layout(title_text="Churn Distribution w.r.t. Internet Service and Gender")

fig.show()

```

Churn Distribution w.r.t. Internet Service and Gender



Business Insights: Internet Service, Gender, and Churn

- **Fiber Optic Churn Remains High:** This detailed visualization confirms that customers with Fiber optic internet service have a significantly higher churn rate (**41.88%** overall for Fiber Optic users) compared to DSL (**18.96%** churn) or no internet service (**7.49%** churn), regardless of gender. This further points to potential dissatisfaction with the service quality or pricing of Fiber optic plans.
- **Gender-Neutral Impact:** The churn patterns across different internet service types are broadly similar for both male and female customers, reinforcing the earlier observation that gender is not a primary driver of churn.
- **No Internet Service = Low Churn:** Customers with no internet service have a very low churn rate (**7.49%**), suggesting they are either highly satisfied with just phone service or have simpler needs that are being met.
- **Actionable Insight:** Prioritize investigating the root causes of churn among Fiber optic customers (e.g., speed consistency, customer support for fiber issues, competitive pricing). Retention efforts should focus on improving the Fiber optic experience.

6.20 Detailed Numerical Feature Analysis - Tenure vs Churn

```

fig = px.box(df, x='Churn', y = 'tenure')

# Update yaxis properties

```

```

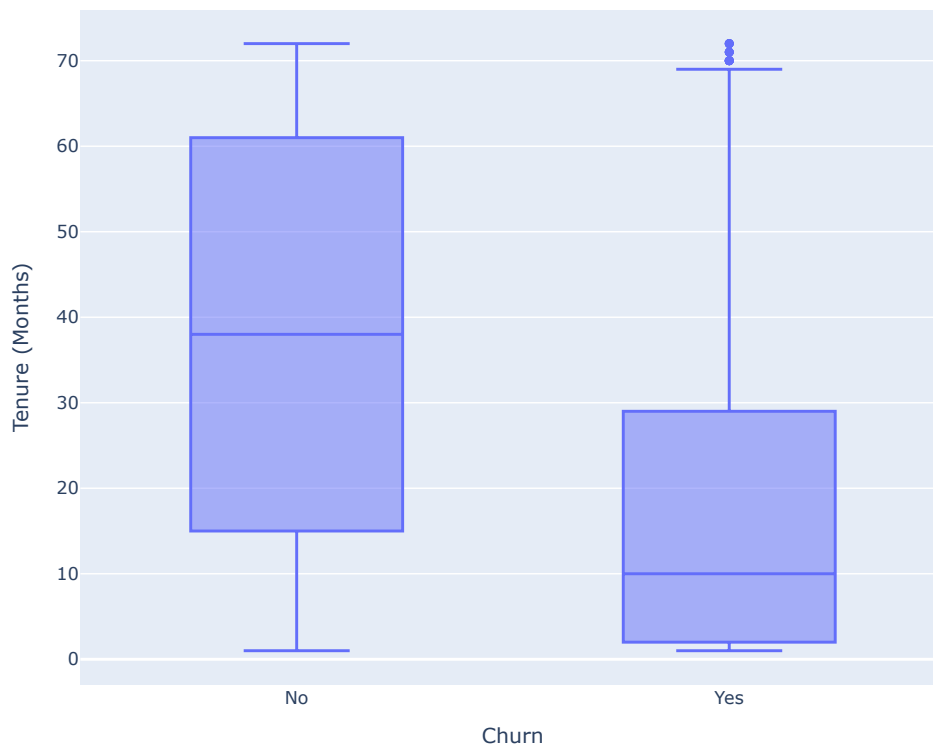
fig.update_yaxes(title_text='Tenure (Months)', row=1, col=1)
# Update xaxis properties
fig.update_xaxes(title_text='Churn', row=1, col=1)

# Update size and title
fig.update_layout(autosize=True, width=750, height=600,
                  title_font=dict(size=25, family='Courier'),
                  title='<b>Tenure vs Churn</b>',
)

fig.show()

```

Tenure vs Churn



Business Insights: Tenure vs Churn (Detailed)

- **New Customer Churn:** The box plot clearly illustrates that customers with very short tenures (median tenure for churned customers is significantly lower) are much more likely to churn. For instance, the median tenure for churning customers is approximately **10 months**, whereas for non-churning customers it's around **38 months**.
- **Long-Term Loyalty:** Conversely, customers with longer tenures are less likely to churn, indicating that once a customer stays beyond a certain period, their loyalty tends to increase.
- **Targeted Onboarding and Early Engagement:** The company should invest heavily in robust onboarding processes and early engagement strategies to ensure new customers quickly find value and feel connected to the service, thereby reducing early churn.

6.21 Detailed Numerical Feature Analysis - Monthly Charges vs Churn

```

sns.set_context("paper", font_scale=1.1)
plt.figure(figsize=(10,6))
ax = sns.kdeplot(df.MonthlyCharges[(df["Churn"] == 'No') ],
                color="Red", shade = True);
ax = sns.kdeplot(df.MonthlyCharges[(df["Churn"] == 'Yes') ],
                ax=ax, color="Blue", shade= True);
ax.legend(["Not Churn","Churn"],loc='upper right');
ax.set_ylabel('Density');
ax.set_xlabel('Monthly Charges');
ax.set_title('Distribution of monthly charges by churn');
plt.show()

```

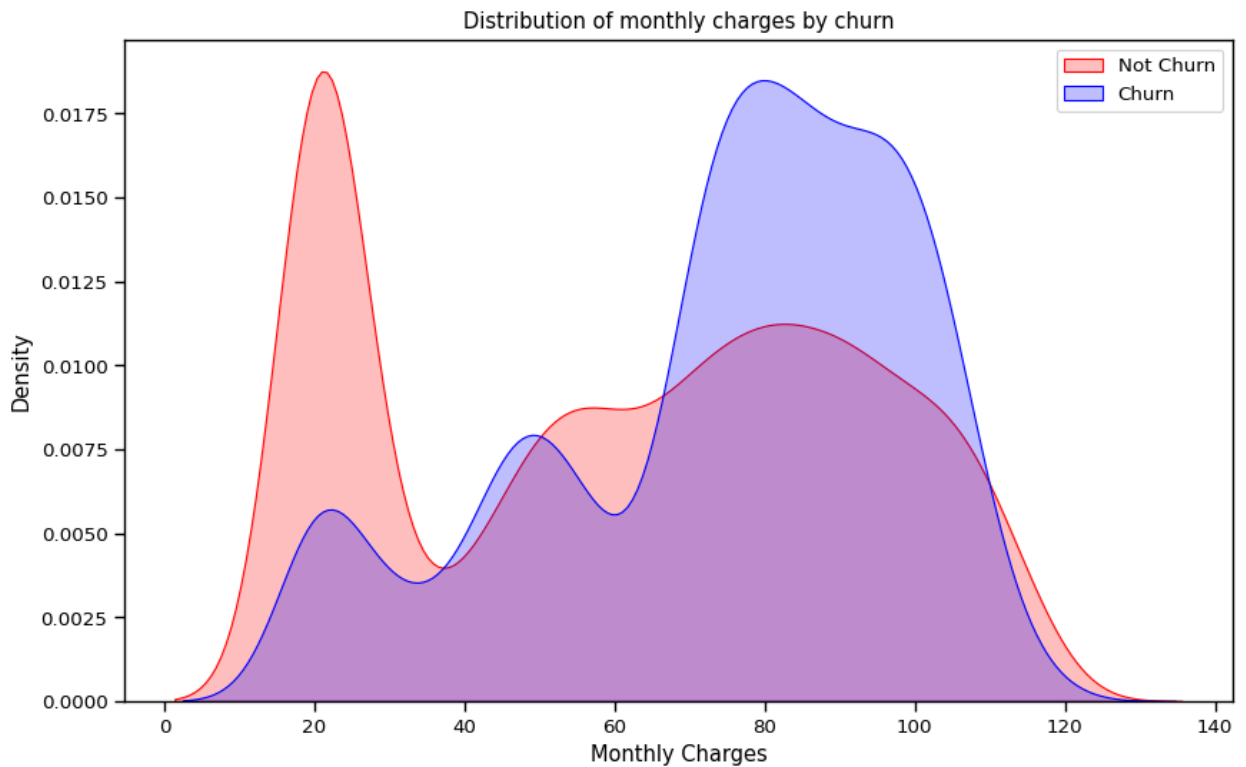


```
/tmp/ipykernel_530/777358235.py:3: FutureWarning:
```

```
`shade` is now deprecated in favor of `fill`; setting `fill=True`.  
This will become an error in seaborn v0.14.0; please update your code.
```

```
/tmp/ipykernel_530/777358235.py:5: FutureWarning:
```

```
`shade` is now deprecated in favor of `fill`; setting `fill=True`.  
This will become an error in seaborn v0.14.0; please update your code.
```



Business Insights: Monthly Charges and Churn (Detailed KDE)

- **High Monthly Charges and Churn:** The KDE plot clearly shows that the density of churning customers peaks at higher monthly charges (around 80—100) compared to non-churning customers, whose distribution is more spread out and has a significant peak at lower charges (around 20—25). This reinforces the idea that customers with expensive plans are more prone to churn.
- **Value Perception:** This suggests that customers paying higher monthly fees might have higher expectations for service quality, features, or support, and if these expectations are not met, they are more likely to churn.
- **Pricing Strategy and Communication:** The company should review its pricing strategy, particularly for higher-tier plans, to ensure perceived value aligns with the cost. Transparent communication about what customers are getting for their money is crucial.

✓ 6.22 Detailed Numerical Feature Analysis - Total Charges vs Churn

```
plt.figure(figsize=(10,6))  
ax = sns.kdeplot(df.TotalCharges[(df["Churn"] == 'No') ],  
                 color="Gold", shade = True);  
ax = sns.kdeplot(df.TotalCharges[(df["Churn"] == 'Yes') ],  
                 ax=ax, color="Green", shade= True);  
ax.legend(["Not Churn", "Churn"], loc='upper right');  
ax.set_ylabel('Density');  
ax.set_xlabel('Total Charges');
```

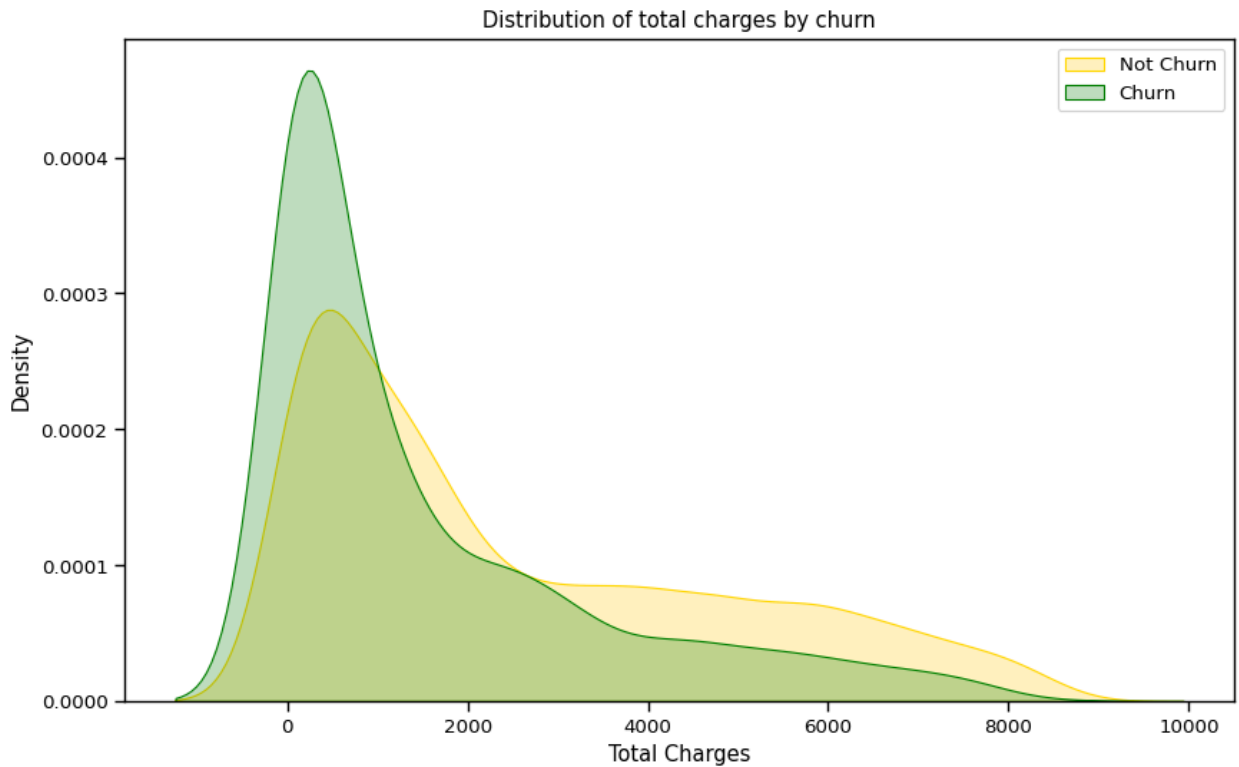
```
ax.set_title('Distribution of total charges by churn');  
plt.show()
```

/tmp/ipykernel_530/1327139003.py:2: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

/tmp/ipykernel_530/1327139003.py:4: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.



Business Insights: Total Charges and Churn (Detailed KDE)

- **Lower Total Charges and Churn:** The KDE plot for Total Charges shows that churning customers tend to have lower total charges, with a higher density concentrated at the lower end of the total charges spectrum (peaking around 100–500). This aligns with our observation that newer customers (who haven't accumulated high total charges) are more likely to churn.
- **Cumulative Value:** Non-churning customers, by contrast, show a more spread-out distribution with higher densities at higher total charges, reflecting their longer tenure and continued use of services, with a significant portion accumulating total charges above \$2000.
- **Early Engagement is Key:** This plot further emphasizes the importance of early customer engagement and value demonstration. Preventing churn in the initial stages of a customer's lifecycle is crucial for increasing their lifetime value.

› 6.23 Correlation Heatmap of All Features

↳ 1 cell hidden

Business Insights: Correlation Heatmap

- **Strong Positive Correlation with Churn:** `Month-to-month contract` (correlation with churn: **0.40**), `InternetService_Fiber optic` (correlation with churn: **0.30**), `PaperlessBilling` (correlation with churn: **0.19**), and `PaymentMethod_Electronic check` (correlation with churn: **0.29**) show moderate positive correlations with `Churn`. This means as these features are present, the likelihood of churn increases. This reinforces our earlier observations from individual feature analysis.
- **Strong Negative Correlation with Churn:** `Tenure` (correlation with churn: **-0.35**), `Contract_Two year` (correlation with churn: **-0.30**), `OnlineSecurity` (correlation with churn: **-0.27**), `TechSupport` (correlation with churn: **-0.28**), and `Contract_One year` (correlation with churn: **-0.18**) show moderate negative correlations with `Churn`. This indicates that customers with longer tenure, longer contracts, and those using security/support services are less likely to churn.
- **Inter-Feature Correlations:** Observe correlations between features themselves (e.g., `tenure` and `TotalCharges` are highly correlated with a value of **0.83**). This can be important for feature selection in modeling, as highly correlated features might not add independent predictive power.

Overall Insight: The heatmap visually summarizes the relationships, confirming many of the individual insights gained from bar plots and box plots. Features with strong positive correlation to churn represent risk factors, while those with strong negative correlation are protective factors.

✓ 7. Data Preprocessing for Machine Learning

Before training a model, we need to preprocess the data. This involves:

- Encoding categorical variables (both binary and multi-class).
- Scaling numerical features (though we might skip this for tree-based models, it's good practice for others).
- Splitting the data into training and testing sets.

✓ 7.1 Encoding Categorical Features

We'll convert all categorical features into numerical representations. Binary categorical features will be mapped to 0 and 1, while multi-class categorical features will be one-hot encoded.

```
# Make a copy of the DataFrame to avoid modifying the original during preprocessing
df_processed = df.copy()

# Map 'Yes'/'No' and 'Male'/'Female' to 1/0 for binary categorical columns
binary_cols = ['Partner', 'Dependents', 'PhoneService', 'PaperlessBilling', 'Churn']
for col in binary_cols:
    df_processed[col] = df_processed[col].map({'Yes': 1, 'No': 0})

df_processed['gender'] = df_processed['gender'].map({'Female': 1, 'Male': 0})

# Handle columns like 'OnlineSecurity', 'OnlineBackup', etc. which have 'No internet service'
# and 'No phone service' categories. We can simplify these first.
service_cols = ['OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
                'StreamingTV', 'StreamingMovies', 'MultipleLines']

for col in service_cols:
    df_processed[col] = df_processed[col].replace({'No internet service': 'No', 'No phone service': 'No'})
    df_processed[col] = df_processed[col].map({'Yes': 1, 'No': 0})

# One-hot encode remaining multi-class categorical columns
multi_class_cols = ['InternetService', 'Contract', 'PaymentMethod']
df_processed = pd.get_dummies(df_processed, columns=multi_class_cols, drop_first=True)

# Display the first few rows of the processed DataFrame and its info to verify
print("First 5 rows of the processed dataset:")
display(df_processed.head())
print("\nProcessed Dataset Info:")
df_processed.info()
```

First 5 rows of the processed dataset:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	OnlineSecurity	OnlineBackup
0	1	0	1	0	1	0	0	0	1
1	0	0	0	0	34	1	0	1	0
2	0	0	0	0	2	1	0	1	1
3	0	0	0	0	45	0	0	1	0
4	1	0	0	0	2	1	0	0	0

5 rows × 24 columns

Processed Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

Index: 7032 entries, 0 to 7042

Data columns (total 24 columns):

#	Column	Non-Null Count	Dtype
0	gender	7032 non-null	int64
1	SeniorCitizen	7032 non-null	int64
2	Partner	7032 non-null	int64
3	Dependents	7032 non-null	int64
4	tenure	7032 non-null	int64
5	PhoneService	7032 non-null	int64
6	MultipleLines	7032 non-null	int64
7	OnlineSecurity	7032 non-null	int64
8	OnlineBackup	7032 non-null	int64
9	DeviceProtection	7032 non-null	int64
10	TechSupport	7032 non-null	int64
11	StreamingTV	7032 non-null	int64
12	StreamingMovies	7032 non-null	int64
13	PaperlessBilling	7032 non-null	int64
14	MonthlyCharges	7032 non-null	float64
15	TotalCharges	7032 non-null	float64
16	Churn	7032 non-null	int64
17	InternetService_Fiber optic	7032 non-null	bool
18	InternetService_No	7032 non-null	bool
19	Contract_One year	7032 non-null	bool
20	Contract_Two year	7032 non-null	bool
21	PaymentMethod_Credit card (automatic)	7032 non-null	bool
22	PaymentMethod_Electronic check	7032 non-null	bool
23	PaymentMethod_Mailed check	7032 non-null	bool

dtypes: bool(7), float64(2), int64(15)

memory usage: 1.0 MB

7.2 Scaling Numerical Features

Numerical features often have different ranges, which can affect the performance of some machine learning algorithms.

We'll use `StandardScaler` to normalize these features.

```
from sklearn.preprocessing import StandardScaler

# Identify numerical columns to scale
numerical_cols_to_scale = ['tenure', 'MonthlyCharges', 'TotalCharges']

# Initialize StandardScaler
scaler = StandardScaler()

# Apply scaling to the numerical columns
df_processed[numerical_cols_to_scale] = scaler.fit_transform(df_processed[numerical_cols_to_scale])

print("Scaled numerical features (first 5 rows):")
display(df_processed[numerical_cols_to_scale].head())
```